

Risk and Return

RECOMMENDED READING:

ESSENTIALS OF MANAGERIAL FINANCE BY SCOTT BESLEY AND EUGENE F. BRIGHAM (CH# 8)

Defining Risk

- Risk refers to the chance of occurring some unfavorable events.
- In finance, precisely, risk is the chance of receiving an actual return other than the expected.
 - variability of returns or outcomes from an investment
 - ✓ i.e., Investment risk

Risk and Probability

- *Probability* - the chance that the event will occur.
- *Probability distribution* - if a probability is assigned to each event after listing down all possible events, or outcomes.

Outcome	Probability
Rain	0.40 = 40%
No rain	$\frac{0.60}{1.00} = \frac{60}{100\%}$

- Probabilities can be assigned to the possible outcomes (or returns) from an investment – i.e., buying a bond; two probable outcomes -
 - i. The issuer makes the interest payments, or
 - ii. The issuer fails to make the interest payments.

The higher the probability of default on the interest payments, the riskier the bond; the higher the risk.

Expected Rate of Return

- Expected rate of return - the mean value of the probability distribution of possible results.
 - Weighted average of the outcomes, with each outcome's *weight being its probability of occurrence*.

Example

Rate of return probability distributions for two companies - Martin Products Inc. and U.S. Electric. under different economic conditions:

TABLE 8-2 Calculation of Expected Rates of Return: Martin Products and U.S. Electric

State of the Economy (1)	Probability of This State Occurring (2)	Martin Products		U.S. Electric	
		Return If This State Occurs (3)	Product: (2) × (3) = (4)	Return If This State Occurs (5)	Product: (2) × (5) = (6)
Boom	0.2	110%	22%	20%	4%
Normal	0.5	22	11	16	8
Recession	0.3	-60	-18	10	3
	<u>1.0</u>		$\hat{r}_{US} = 15\%$		$\hat{r}_{US} = 15\%$

Calculating Expected Rate of Return

- Expected return is designated with $\hat{r}$ (r- hat)
 - Indicate the return expected or is uncertain about the possible outcomes in the future.

Expected rate of return ($\hat{r}$) = $Pr_1r_1 + Pr_2r_2 + \dots + Pr_nr_n$

$$= \sum_{i=1}^n Pr_i r_i$$

Calculation of ERR from the example in previous slide (Slide#4)

$$\begin{aligned}\text{Expected rate of return of Martin Inc. } (\hat{r}_M) &= Pr_1r_1 + Pr_2r_2 + Pr_3r_3 \\ &= 0.20 * (110\%) + 0.50 * (22\%) + 0.30 * (-60\%) \\ &= 22\% + 11\% - 18\% = 15\%\end{aligned}$$

$$\begin{aligned}\text{Expected rate of return of US Electric. } (\hat{r}_U) &= Pr_1r_1 + Pr_2r_2 + Pr_3r_3 \\ &= 0.20 * (20\%) + 0.50 * (16\%) + 0.30 * (10\%) \\ &= 4\% + 8\% + 3\% = 15\%\end{aligned}$$

Calculating Expected Rate of Return

Problem

Suppose you have predicted the following returns for stocks C and T in three possible states of nature. What are the expected returns?

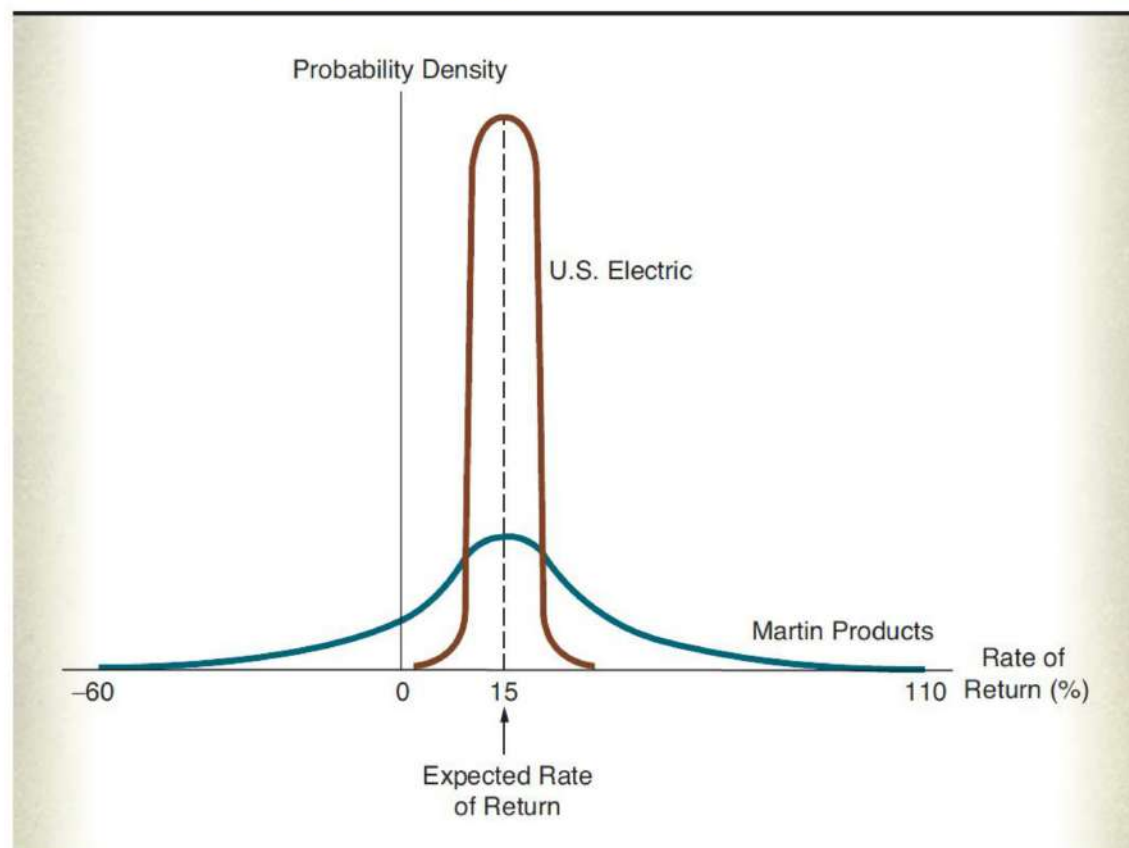
State	Probability		C	T
Boom	0.3	✕	0.15	0.25
Normal	0.5	✕	0.10	0.20
Recession	0.2		0.02	0.01

Solution

$$\hat{r}_C = 0.3 (0.15) + 0.5(0.10) + 0.2(0.02) = 0.099 = 9.9\%$$

$$\hat{r}_T = 0.3(0.25) + 0.5(0.20) + 0.2(0.01) = .177 = 17.7\%$$

FIGURE 8-2 Continuous Probability Distributions of Martin Products' and U.S. Electric's Rates of Return



Risk and Return Relationship

- The tighter the probability distribution, the lower the risk.
 - The tighter the probability distribution
 - the less variability - less likely that the actual return will differ dramatically from the expected return.
 - ✓ US Electric's return is more stable compared to Martin Inc.

Measuring Risk [1/4]

- The width of a probability distribution indicates the variability of the possible outcomes.
- The most commonly used measure of risk is *standard deviation* (σ)
 - The smaller the standard deviation (SD), the tighter the probability distribution
 - Accordingly, the lower the total risk (associated with the investment)
- Another measure of risk is called variance, σ^2
 - Helpful to calculate the absolute value of deviation of individual returns from expected return.
 - ✓ i.e., $(2)^2 = 4$ and $(-2)^2 = 4$
- Relationship between SD and Variance

$$\text{SD } (\sigma) = \sqrt{\sigma^2}$$

Measuring Risk [2/4]

- If we have the expected return ($\hat{r}$), then we can calculate the variance as follows:

$$\begin{aligned}\text{Variance } (\sigma^2) &= (r_1 - \hat{r})^2 Pr_1 + (r_2 - \hat{r})^2 Pr_2 + \dots + (r_n - \hat{r})^2 Pr_n \\ &= \sum_{i=1}^n (r_i - \hat{r})^2 * Pr_i\end{aligned}$$

Where,

$r_1, r_2, \dots, r_n$ are the individual returns.

$Pr_1, Pr_2, \dots, Pr_n$ are the individual probabilities that may take place.

$$\text{So, the SD } (\sigma) = \sqrt{\sum_{i=1}^n (r_i - \hat{r})^2 * Pr_i}$$

Measuring Risk [3/4]

Calculating risk: Example (from Slide#4)

TABLE 8-3 Calculating Martin Products' Standard Deviation

Payoff r_i (1)		Expected Return $\hat{r}$ (2)		Deviation $r_j - \hat{r}$ (1) - (2) = (3)	$(r_j - \hat{r})^2 =$ (4)	Probability (5)	$(r_j - \hat{r})^2 \text{Pr}_i$ (4) $\times$ (5)	=	σ^2 (6)
110%	–	15%	=	95	9,025	0.2	$9,025 \times 0.2$	=	1,805.0
22	–	15	=	7	49	0.5	49×0.5	=	24.5
–60	–	15	=	–75	5,625	0.3	$5,625 \times 0.3$	=	1,687.5
Variance = σ^2								=	<u>3,517.0</u>

$$\text{Standard deviation} = \sigma = \sqrt{\sigma^2} = \sqrt{3,517} = 59.3\%$$

Similarly, the SD of US Electric. = 3.6%
(Please calculate following the steps in Table 8-3).

Measuring Risk [4/4]

Problem

Suppose you have predicted the following returns for stocks C and T in three possible states of nature. Calculate the risk of each stock if the expected return of stocks C: 9.90% and T: 17.70%. Which stock is more risky?

State	Probability	C	T
Boom	0.30	0.15	0.25
Normal	0.50	0.10	0.20
Recession	0.20	0.02	0.01

Solution

$$\begin{aligned}
 \sigma_C &= \sqrt{(r_1 - \hat{r})^2 Pr_1 + (r_2 - \hat{r})^2 Pr_2 + (r_3 - \hat{r})^2 Pr_3} \\
 &= \sqrt{(0.15 - 0.099)^2 * 0.30 + (0.10 - 0.099)^2 * 0.50 + (0.02 - 0.099)^2 * 0.20} \\
 &= \sqrt{.002029} = 0.045 = 4.50\%
 \end{aligned}$$

$$\begin{aligned}
 \sigma_T &= \sqrt{(0.25 - 0.177)^2 * 0.30 + (0.20 - 0.177)^2 * 0.50 + (0.01 - 0.177)^2 * 0.20} \\
 &= \sqrt{0.007441} = 0.0863 = 8.63\%
 \end{aligned}$$

Stock T is more risky than stock C (so does its return: [Slide#6](#)).

Measuring Risk and Return for Unknown Probability Distribution [1/2]

- In many cases, the actual probability distributions are not known (unlike the previous cases)
 - We need to calculate return from available data over some past period.
 - Using past information to predict future risk and return.
- In such scenarios, the formulae for return and risk calculations are as follows:

$$\text{Return, } \bar{r} = \frac{\sum_{t=1}^n \ddot{r}_t}{n}$$

Where,

$\bar{r}$ = arithmetic average of the annual returns earned during the last n years or mean return.

n = number of periods

$$\text{Risk, } (\sigma \text{ or } s) = \sqrt{\frac{\sum_{t=1}^n (\ddot{r}_t - \bar{r})^2}{n-1}}$$

Where,

$\ddot{r}_t$ represents the past realized rate of return in Period t

Measuring Risk and Return for Unknown Probability Distribution [2/2]

Example

we have observed the following returns associated with a common stock X:

Year	$\hat{r}$
2008	15%
2009	-5
2010	20
2011	22

Calculate the risk and return of stock X.

Solution

$$\text{Return } (\bar{r}) = \frac{15 + (-5) + 20 + 22}{4} = 13\%$$

$$\text{Estimated Risk } (\sigma) = \sqrt{\frac{(15-13)^2 + (-5-13)^2 + (20-13)^2 + (22-13)^2}{4-1}} = \sqrt{\frac{458}{3}} = 12.4\%$$

Coefficient of Variation (Risk/Return Ratio)

- A standardized measure of the risk per unit of return.

$$\text{Coefficient of Variation (CV)} = \frac{\text{Risk}}{\text{Return}} = \frac{\sigma}{\hat{r}}$$

- It provides a more meaningful basis for comparison of risks and returns of alternative investments
 - ✓ i.e., risk of Martin Inc. (59.3%) differ from US Electric (3.6%), although they have same expected return (15%).
 - ✓ Here, since returns are same, we can clearly see Martin Inc. is riskier than US Electric as it has higher SD.
- The calculation of CV would give us a compare risks in relation to their returns

$$\text{Therefore, } CV_M = \frac{59.3\%}{15\%} = 3.95; CV_U = \frac{3.6\%}{15\%} = 0.24$$

Hence, Martin is approx. 16.5X (3.95/0.24) riskier than US Electric. to generate 1 unit of return.

- *Higher the CV, higher the risk (for a given level of return)*
- *However, calculating CV is more beneficial to compare risk and return of alternative projects/ investment when the risk and return values are unequal.*

Calculating Coefficient of Variation (CV)

Problem

Following is the information for two stocks

Investment	Expected Return	Standard Deviation
Stock D	10%	8%
Stock E	36%	24%

Which investment has the greater relative risk?

Solution

$$CV_D = \frac{\sigma}{\hat{r}} = \frac{0.08}{0.10} = 0.8 = 80\%$$

$$CV_E = \frac{0.24}{0.36} = 0.667 = 66.70\%$$

Stock D is relatively riskier than Stock E.

Portfolio – the Concept

- A portfolio is a collection of (financial) assets
 - Diversifying investment in different assets (i.e., stocks, bonds, or other assets) instead of investing only in one.
- A portfolio is a key tool for managing risk and maximizing returns in the world of finance.
- By holding a range of investments,
 - the risk of losing money on any one investment is reduced
 - Also, gains from some investments may offset losses from others
- A common phrase associated with portfolio management is

Don't put all your eggs in one basket

Portfolio Return [1/2]

The expected return on a portfolio ($\hat{r}_p$), is simply the weighted average of the expected returns on the individual assets in the portfolio.

$$\hat{r}_p = w_1 \hat{r}_1 + w_2 \hat{r}_2 + \dots + w_N \hat{r}_N = \sum_{j=1}^N w_j \hat{r}_j$$

Where,

the portfolio includes N assets.

w_1 = the weights or proportion of individual assets in the portfolio (the sum is 1)

$\hat{r}_1$ = the expected returns on the individual assets in the portfolio

Portfolio Return [2/2]

Example

Suppose security analysts estimate that the following returns could be expected on four large companies:

Company	Expected Return, $\hat{r}$
AT&T	8%
Citigroup	13
General Electric	19
Microsoft	16

If you have \$100,000 and want to invest equally in the four stocks, what would be the expected return of the portfolio investment?

Solution

$$\begin{aligned}\hat{r}_p &= w_{ATT}\hat{r}_{ATT} + w_{Citi}\hat{r}_{Citi} + w_{GE}\hat{r}_{GE} + w_{MS}\hat{r}_{MS} \\ &= 0.25 (8\%) + 0.25 (13\%) + 0.25 (19\%) + 0.25 (16\%) = 14\%\end{aligned}$$

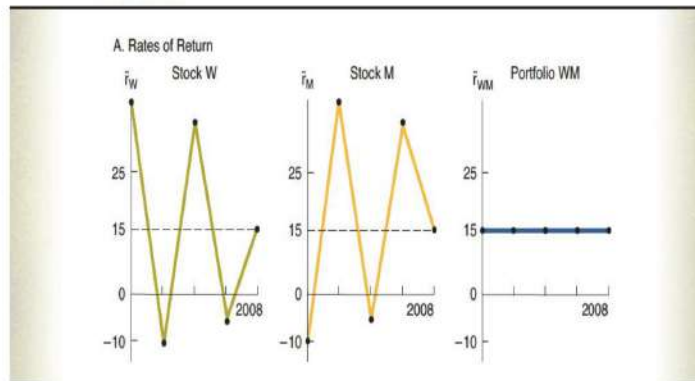
Portfolio Risk [1/6]

- Unlike returns, a portfolio risk (σ_p) is *not a weighted average of the standard deviations* of the individual securities in the portfolio.
- The purpose of constructing portfolio is to minimize risks in the investment even if the individual assets in the portfolio is risky.

Portfolio Risk [2/6]

- Portfolio risk range is usually expressed as a range of correlation values between negative and positive numbers, expressed in correlation coefficient (ρ): - 1 to +1
 - If $\rho = -1$: perfect negative correlation between the assets in the portfolio
 - Reduce the overall risk of the portfolio
 - ✓ when one asset increases in value, the other asset decreases in value by the same proportion and vice versa
 - If $\rho = +1$: a perfect positive correlation between the assets in the portfolio
 - Increase the overall risk of the portfolio
 - ✓ when one asset increases in value, the other asset also increases in value by the same proportion and vice versa
 - If $\rho = 0$: no correlation between the assets in the portfolio
 - No impact on portfolio risk diversification
 - ✓ movements of one asset are not related to the movements of the other asset

FIGURE 8-5 Rate of Return Distribution for Two Perfectly Negatively Correlated Stocks ($\rho = -1.0$) and for Portfolio WM



Year	Stock W ($\bar{r}_W$)	Stock M ($\bar{r}_M$)	Portfolio WM ($\bar{r}_{WM}$)
2004	40%	-10%	15%
2005	-10	40	15
2006	35	-5	15
2007	-5	35	15
2008	15	15	15
Average return, $\bar{r}$	15%	15%	15%
Standard deviation, s	22.6%	22.6%	0%

Portfolio Risk [3/6]

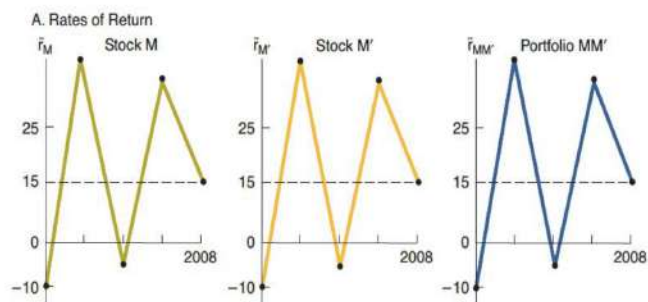
Example (Perfectly negative correlation, $\rho = -1$)

To construct Portfolio WM, 50 percent of the total amount invested is invested in Stock W and 50 percent is invested in Stock M.

From the Figures,

- The two stocks would be quite risky if they were held in isolation.
- Stocks W and M can be combined to form a risk-free portfolio is because their returns move in opposite directions

FIGURE 8-6 Rate of Return Distributions for Two Perfectly Positively Correlated Stocks ($\rho = +1.0$) and for Portfolio MM'



Portfolio Risk [4/6]

Example (Perfectly positive correlation, $\rho = +1$)

From the Figures,

- The combination of Stocks W and M in a portfolio does not eliminate risk because their returns move in same directions

Year	Stock M ($\bar{r}_M$)	Stock M' ($\bar{r}_{M'}$)	Portfolio MM' ($\bar{r}_{MM'}$)
2004	-10%	-10%	-10%
2005	40	40	40
2006	-5	-5	-5
2007	35	35	35
2008	15	15	15
Average return, $\bar{r}$	15%	15%	15%
Standard deviation, s	22.6%	22.6%	22.6%

Portfolio Risk [5/6]

- Other than perfectly negative or positive correlations –
 - Risk can be reduced if the correlation between assets in the portfolio is negative.
 - ✓ i.e., -0.10, -0.50, -0.90 etc.: the higher the negative value, lower is the risk of the portfolio.
 - Risk will increase if the correlation between assets in the portfolio is positive.
 - ✓ i.e., +0.10, +0.50, +0.90 etc.: the higher the positive value, higher is the risk of the portfolio.

Hence, the portfolio risk calculation requires the correlation coefficient of the assets in the portfolio.

Accordingly, the risk of a portfolio consisting two assets can be calculated as follows:

$$\text{Portfolio Variance } \sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{1,2} \sigma_1 \sigma_2$$

$$\text{Portfolio SD, } \sigma_p = \sqrt{\sigma_p^2} = \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{1,2} \sigma_1 \sigma_2}$$

Where,

w_1 = Weight of Asset 1 in the portfolio

w_2 = Weight of Asset 2 in the portfolio

σ_1 = SD of Asset 1 in the portfolio

σ_2 = SD of Asset 2 in the portfolio

$\rho_{1,2}$ = Correlation Coefficient of Assets 1 & 2.

Portfolio Risk [6/6]

Problem

Suppose you have invested in two stocks, A and B, with the following characteristics:

Stock A has an expected return of 12% and a standard deviation of 20%.

Stock B has an expected return of 8% and a standard deviation of 15%.

The correlation coefficient between the returns of Stock A and Stock B is 0.40.

You have invested 60% of your portfolio in Stock A and 40% in Stock B. What is the risk of your portfolio?

Solution

$$\begin{aligned}\text{Portfolio SD, } \sigma_p &= \sqrt{0.60^2 * 0.20^2 + 0.40^2 * 0.15^2 + 2 * 0.60 * 0.40 * 0.20 * 0.15 * 0.40} \\ &= \sqrt{0.0144 + 0.0036 + 0.0058} \\ &= \sqrt{0.0238} = 0.15427 = 15.43\%\end{aligned}$$

Where,

$$w_A = 0.60$$

$$w_B = 0.40$$

$$\sigma_A = 0.20$$

$$\sigma_B = 0.15$$

$$\rho_{A,B} = 0.40$$

$$\sigma_p = ?$$

Portfolio Risk - Diversifiable vs. Non-diversifiable

Diversifiable Risk

- Part of a security's risk associated with outcomes generated by events, or behaviors, specific to a firm; i.e., strikes, successful/ unsuccessful marketing programs, etc.
 - Can be eliminated (or reduced) by proper diversification; i.e., constructing portfolio.
- It is specific to a firm characteristics; does not originate from system or market
 - It is called *firm-specific* or *unsystematic risk*.

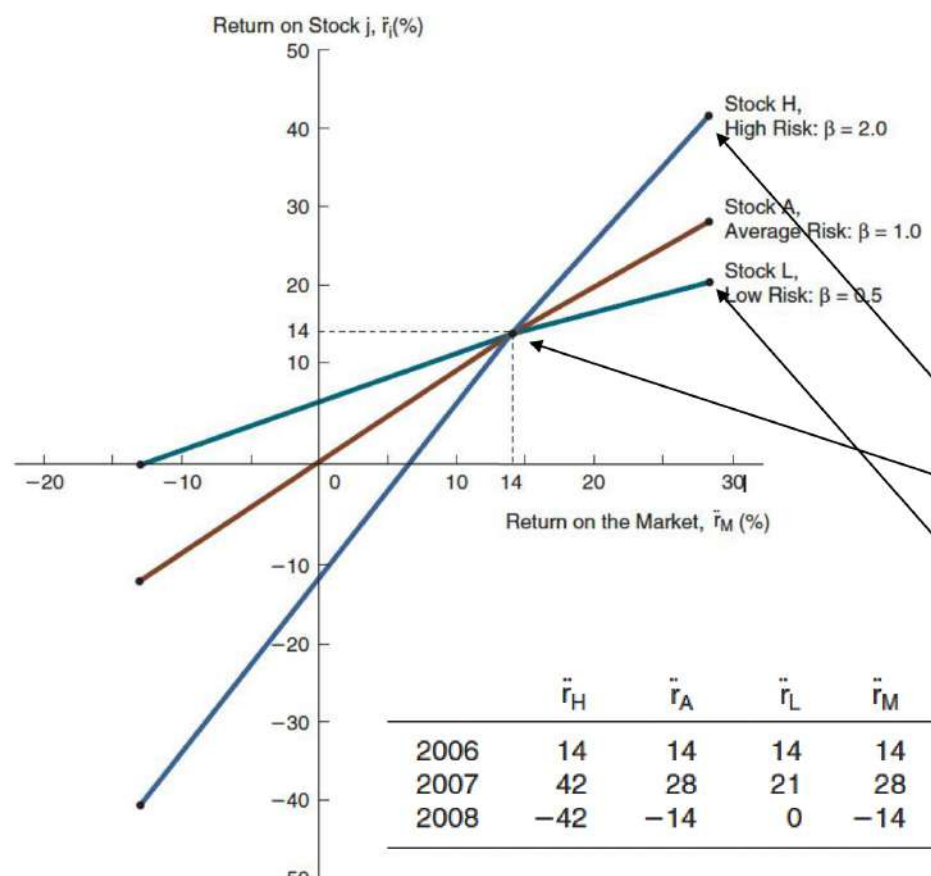
Non-diversifiable Risk

- Part of a security's risk associated with economic or market factors that systematically affect firms.
 - Affect all firms in the system/market; thus called, *systematic* or *market risk*. i.e., *inflation*, *Interest rate change*, *war*, etc.
 - Cannot be eliminated by portfolio diversification

Market Risk and Beta (β)

- Total risk = Diversifiable Risk + Non-diversifiable Risk
- Relevant risk of a security risk that cannot be diversified away
 - The risk that remains after diversifying – the security's market risk (Non-diversifiable risk).
- The coefficient that measures the extent to which the returns on a given stock move with the stock market is called Beta (β).
 - The general movements in the financial markets (i.e., NYSE, DSE) reflect movements in the economy
 - So, we can measure the market risk of a stock by observing its tendency to move with the market

FIGURE 8-9 Relative Volatility of Stocks H, A, and L



Market Risk and Beta (β)

- Beta (β) can also be explained as the slope coefficient of a “regression line” between market return and individual stock return.

For a stock,

- $\beta = 1$ means it has the same level of risk of the market or the average risk of all stocks in the market
- $\beta > 1$ means it has a relatively greater level of risk compared to the average market risk
- $\beta < 1$ means it has a relatively lower level of risk compared to the average market risk

Market Risk, Firm-Specific Risk and Beta (β)

- A stock's risk consists of two components: market risk (systematic/ non-diversifiable risk) and firm-specific risk (unsystematic/diversifiable risk).
- Firm-specific risk can be eliminated through diversification.
- Since unsystematic risk is diversifiable through constructing portfolios, systematic risk is the only risk that is relevant to a rational, diversified investor.
- The market (systematic) risk of a stock is measured by its beta coefficient
- Since a stock's beta coefficient determines how the stock affects the riskiness of a diversified portfolio,
 - beta is a better measure of a stock's relevant risk than is standard deviation, which measures total risk.
- Investors must be compensated for bearing (systematic) risk
 - That is, the greater the riskiness of a stock (i.e., higher β), the higher its required return.

Portfolio Beta

- It is a weighted average of the individual securities' betas included in the portfolio.
 - A measure of the systematic risk of a portfolio of investments in relation to the overall market

$$\begin{aligned}\text{Portfolio Beta } \beta_p &= w_1 \beta_1 + w_2 \beta_2 + \dots + w_N \beta_N \\ &= \sum_{j=1}^N w_j \beta_j\end{aligned}$$

Where,

w_j is the fraction/weight of the portfolio invested in the j^{th} stock

β_j is the beta coefficient of the j^{th} stock.

Portfolio Beta – Example

Problem

Consider the following four securities

Security	Weight	Beta
DCLK	0.13	4.03
KO	0.20	0.84
INTEC	0.27	1.05
KEI	0.40	0.59

What is the portfolio beta?

Solution

$$\begin{aligned}\beta_p &= \sum_{j=1}^{N=4} w_j \beta_j \\ &= w_{DCLK} \beta_{DCLK} + w_{KO} \beta_{KO} + w_{INTEC} \beta_{INTEC} + w_{KEI} \beta_{KEI} \\ &= .13 (4.03) + .2(.84) + .27(1.05) + .4(.59) = 1.22\end{aligned}$$

Relationship Between Risk, Risk Premium and Return – CAPM

CAPM – Capital Asset Pricing Model

A model used to determine the required return on an asset (i.e., stock) considering two things –

1. Risk-free return (R_f)
 - The rate of return an investor can earn without taking any risk
 - ✓ i.e., return from 3-month treasury bill issued by BB
2. Risk premiums
 - *Market risk premium* (R_{pm}) - the additional return over the risk-free rate needed to compensate investors for assuming an average amount of risk by investing the (stock) market.
 - *Stock risk premium* - the additional return to compensate investors risk for investing in a particular stock
 - Product of market risk premium (R_{pm}) and beta of an individual stock β_j : $R_{pm} * \beta_j$

Relationship Between Risk, Risk Premium and Return – CAPM

Therefore, as per CAPM, a stock's required rate of return is –

$$\begin{aligned} r_j &= R_f + \underbrace{(R_m - R_f)}_{R_{pm}} * \beta_j \\ &= R_f + R_{pm} * \beta_j \end{aligned}$$

Where,

R_f = Risk-free rate of return

R_m = Market return – return from a portfolio consisting of all stocks in the stock market.

$R_m - R_f = R_{pm}$ = Market risk premium

β_j = Sensitivity of individual stock to the market movement

Calculating Stock Return Using CAPM

Problem 1

A stock has a beta of 1.25, the expected return on the market is 11.7 percent, and the risk-free rate is 3.5 percent. What must the expected return on this stock be?

Solution

$$\begin{aligned} r_j &= R_f + (R_m - R_f) * \beta_j \\ &= 0.0350 + (0.1170 - 0.0350) * 1.25 \\ &= 0.1375 = 13.75\% \end{aligned}$$

Here,

$$R_f = 3.50\% \text{ or } 0.0350$$

$$R_m = 11.70\% \text{ or } 0.1170$$

$$\beta_j = 1.25$$

$$r_j = ?$$

Calculating Stock Return Using CAPM

Problem 2

Consider the beta of two companies: 1.05 (INTEC) and 0.59 (KEI). If the risk-free rate is 3.15% and the market risk premium is 9.50%, what is the expected return for each? Which stock is riskier? How do you interpret the required return of the stocks in relation to their risk

Solution

$$r_{INTEC} = 0.0315 + 0.0950 * 1.05 = 0.13125 = 13.13\%$$

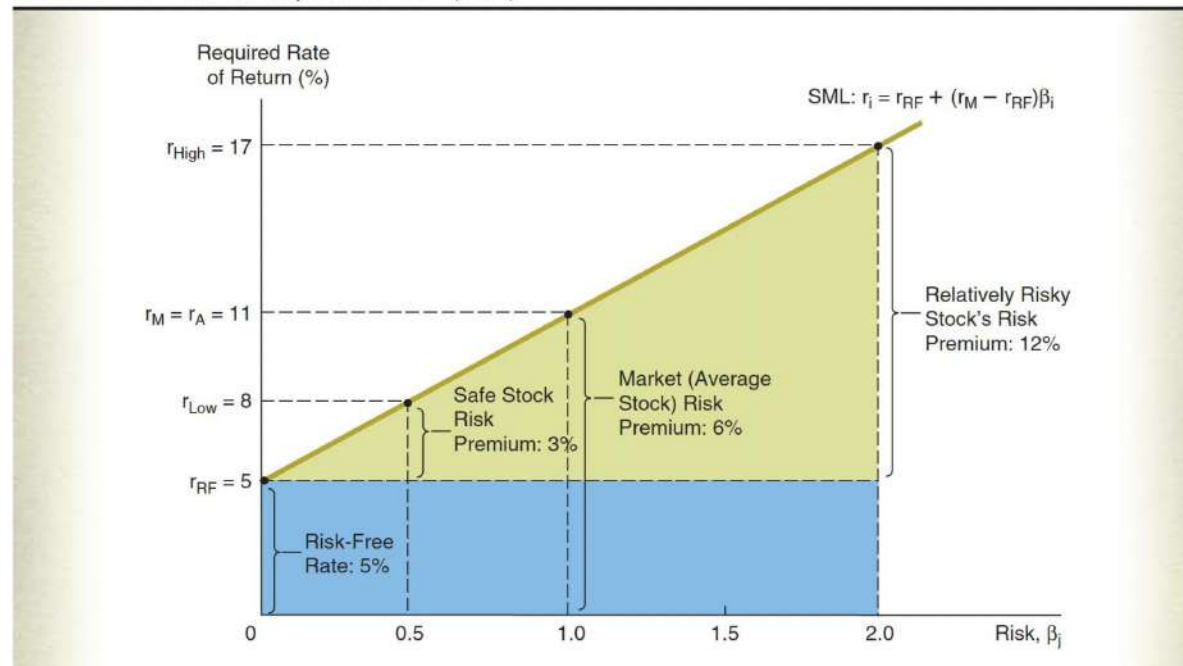
$$r_{KEI} = 0.0315 + 0.0950 * 0.59 = 0.08755 = 8.76\%$$

INTC stock is riskier than KEI stock ($\beta_{INTC}: 1.05 > \beta_{KEI}: 0.59$).

Also, INTEC stock is more sensitive to market movement ($\beta_{INTC}: 1.05 > \beta_M: 1$), so requires more premium (9.98%) than market (9.50%). Thus, required return of INTC stock is higher than the average return of market.

In contrast, KEI stock is less sensitive to market movement ($\beta_{KEI}: 0.59 < \beta_M: 1$), so requires less premium (5.61%) than market (9.50%). Thus, required return of KEI stock is lower than the average return of market.

FIGURE 8-10 The Security Market Line (SML)



CAPM and Security Market Line (SML)

- The graphical illustration of CAPM is Security Market Line (SML)
 - The line that shows the relationship between risk as measured by
 - ✓ beta and
 - ✓ the required rate of return for individual securities.